

Hardware Components Identification -- Guided Notes

ANSWER KEY

Unit 2.1 | CompTIA Network+ N10-009 Obj. Core 1 3.4 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

LGA (Land Grid Array) sockets have _____ on the motherboard socket itself, while the CPU has flat _____. PGA (Pin Grid Array) sockets have pins on the _____ and the socket has _____. Intel uses _____ sockets; AMD traditionally used _____ sockets (though AM5 is LGA).

Answer: Pins; contact pads; CPU; holes. Intel = LGA; AMD traditionally = PGA (AM4 and earlier).

Question 2

On a motherboard, a PCIe x16 slot is _____ than a PCIe x1 slot. An x16 slot is used primarily for _____. A PCIe x1 slot is used for smaller cards such as _____ and _____. A graphics card designed for an x16 slot will/will not fit in an x1 slot.

Answer: Longer; discrete graphics cards (GPUs); sound cards; network cards (any expansion cards). Will NOT fit -- the x16 card is physically longer.

Question 3

An M.2 slot on a motherboard is a small _____ -shaped connector on the board surface. It accepts an M.2 SSD that mounts flat against the board and is secured with a single _____. A SATA port, by contrast, is an _____ -shaped connector used with a cable to connect 2.5-inch and 3.5-inch drives.

Answer: Rectangular (edge); screw; L-shaped (right-angle / small rectangular).

Question 4

The 24-pin ATX connector supplies _____ from the power supply to the motherboard. The small headers at the front edge of the motherboard connect to _____ panel cables. These cables control the _____ button, _____ LED, and _____ LED.

Answer: Main power (voltage and current); front panel; power; HDD activity; reset.

Question 5

SO-DIMM (Small Outline DIMM) RAM modules are used in _____ and are physically _____ than full-size DIMMs used in desktops. A standard desktop DIMM is approximately _____ mm long, while a SO-DIMM is approximately _____ mm long.

Answer: Laptops (and small form factor systems); shorter; 133 mm; 67 mm.

Question 6

A system with _____ graphics uses a GPU built into the CPU or motherboard chipset. Display output ports in this case are located on the _____. A _____ GPU is a separate add-in card. When a discrete GPU is installed, display cables should be connected to the _____, not the motherboard.

Answer: Integrated; motherboard (rear I/O panel); discrete; GPU (graphics card).

Question 7

High-end discrete GPUs require supplemental power from the PSU via _____ connectors (6-pin or 8-pin PCIe power connectors). These connectors supply the _____ V rail directly to the GPU. If these connectors are not plugged in, the system may _____ or the GPU may not be detected.

Answer: PCIe power (auxiliary power); +12V; fail to POST (not power on / produce no display).

Question 8 -- Real World Application

REAL WORLD: A student opens a used desktop tower to identify its components before upgrading it. They see a short rectangular connector on the surface of the motherboard, a long connector on the motherboard near a large fan, two different-length slots on the motherboard, and a cable going to the drive labeled SATA. Identify each component and explain the function of two of them in detail.

Answer: Short rectangular connector on the board surface = M.2 slot (for M.2 NVMe or SATA SSDs). Long connector on the motherboard near a large fan = CPU socket (holds the processor). Two different-length motherboard slots = PCIe x16 (for GPU) and PCIe x1 (for expansion cards). Cable labeled SATA = SATA data cable connecting to a hard drive or optical drive. The M.2 slot connects directly to the PCIe bus, bypassing SATA entirely for NVMe speeds. The PCIe x16 slot receives the GPU and feeds it +12V power plus data bandwidth.